*Machine Learning for Tackling Climate Change*

# Cross-Basin Generalisation for Tropical Cyclone Forecasting using TropiCycloneNet

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*ELEC70122 | Imperial College London | March 27, 2026*

# Motivation: Why Cross-Basin Transfer?

**1 billion**
people live in areas affected by tropical cyclones[1]

**USD 2.5 trillion**
in damages caused by tropical cyclones since 2000[1]

**Our Objective:** to investigate what **cross-basin transfer learning** can achieve for cyclone direction and intensity classification when **data is limited**

**Our Approach:**

Configure existing **DL architectures** — Train on **data-rich region** — **Zero-shot** directly on **data scarce-region test set**

Train on **data-rich region** — **Fine tune** on a **small data-scarce region set** — Evaluate on **data scarce-region test set**

# Why Western Pacific and South Pacific Basins?

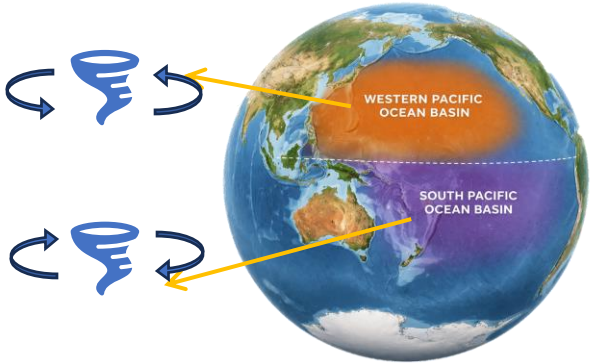
**TropiCycloneNet²**

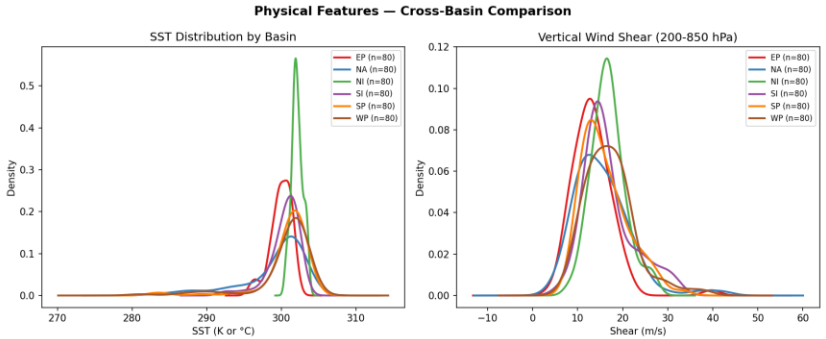
**Transfer pair:**

- ✓ **South Pacific (SP):** 30 storms, 922 timesteps
- ✓ **Western Pacific (WP):** 131 storms, 4,500+ timestamps

**Key classification tasks:**

- ✓ **Direction:** storms **curve** in **opposite** directions across hemispheres, so a WP-trained model learns the wrong movement patterns for SP
- ✓ **Intensity:** driven by **sea surface temperature** & **wind shear**, which overlap across both basins[3], so the physics should transfer




**Our Hypothesis:** intensity prediction should transfer better than direction prediction, because the underlying physics appears to be shared while direction patterns are hemisphere-dependent.

# Data & Processing

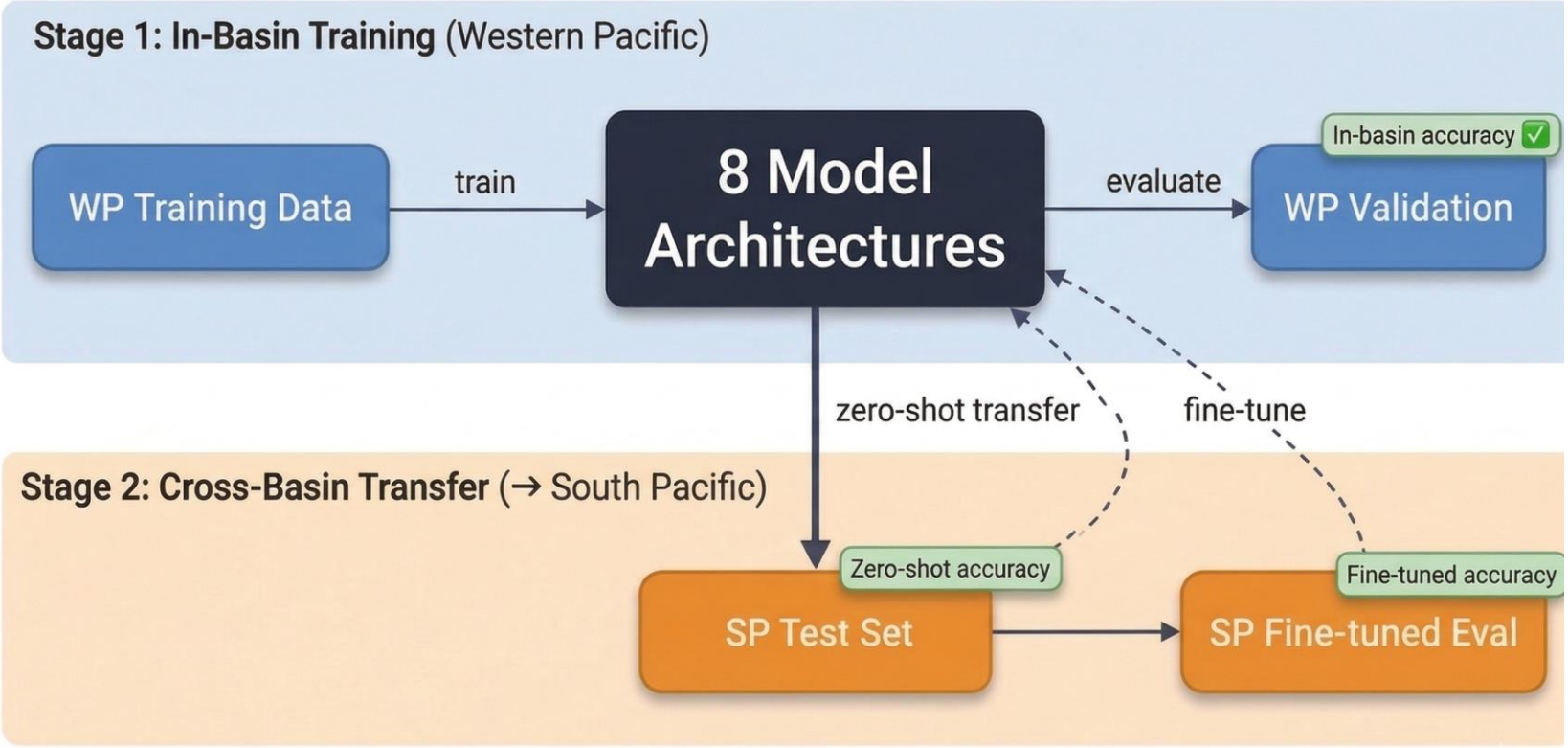
**Data:** Spatial snapshot of the **surrounding atmosphere** + Storm's current **position** & **strength** + Summary **environmental conditions**

**Processing:**

- **Basin-leaking removal:** Stripped environmental features encoding basin identity
- **Hemisphere-aware transforms:** Mirrored SP direction labels & flipped the N-S wind component
- **Derived physics features:** Added wind shear & low-level vorticity
- **Normalisation:** All normalisation computed from WP training data only (no SP leakage)
- **Storm-level splitting:** Split by entire storms rather than individual timesteps

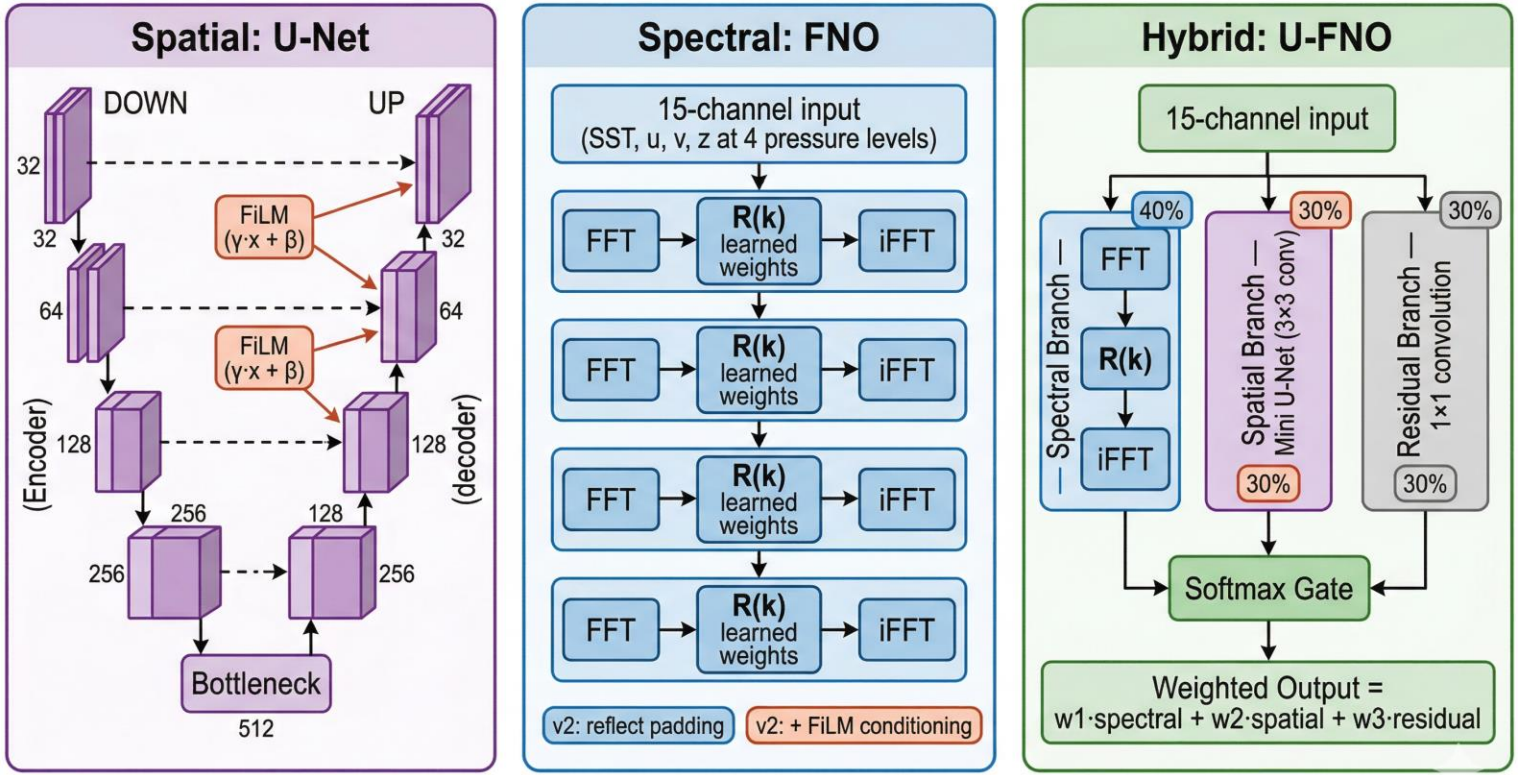
# Method

# Two-Stage Pipeline




***Key question***
*Does fine-tuning always help?*

# Spatial vs Spectral Families

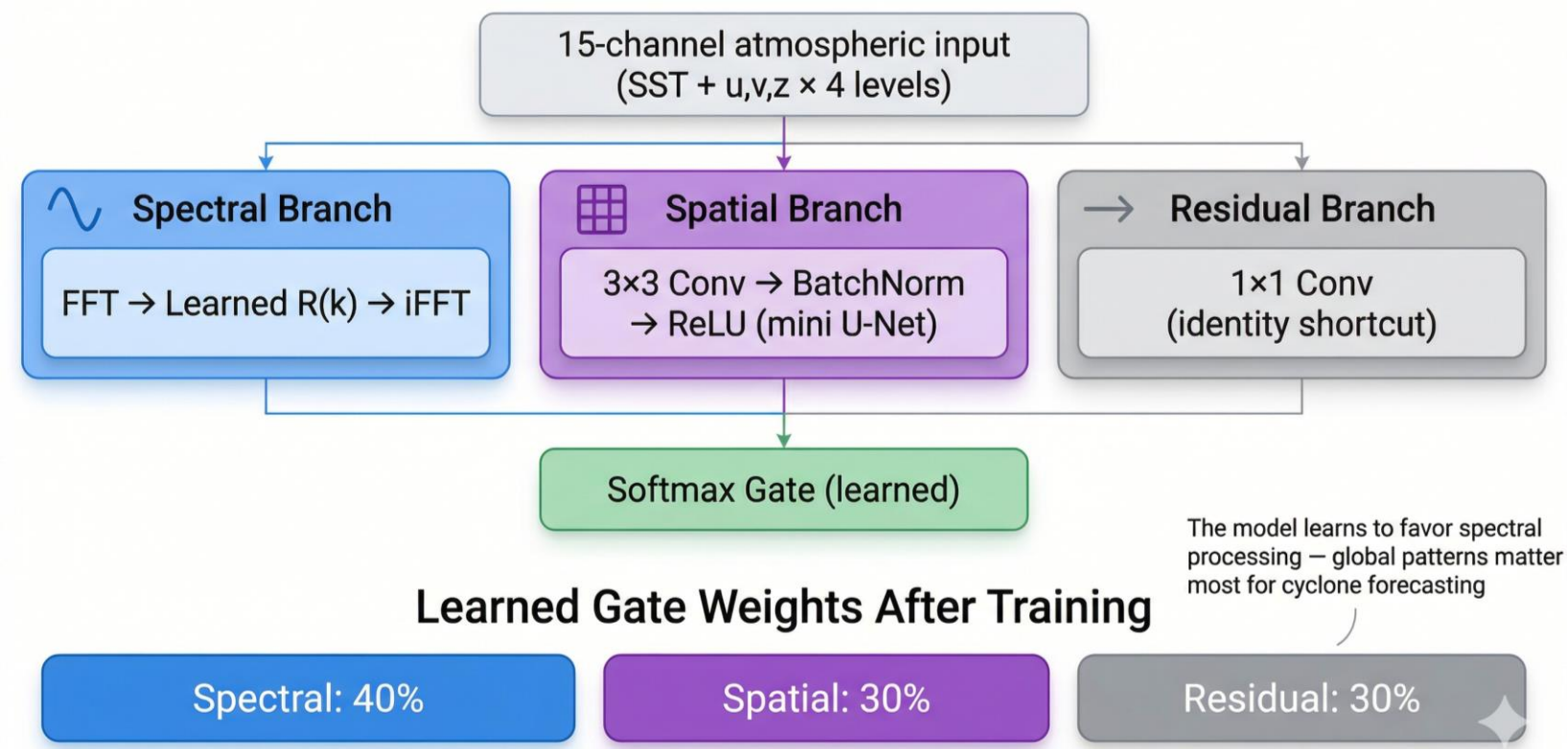



Captures local spatial wind patterns

Global receptive field in every layer

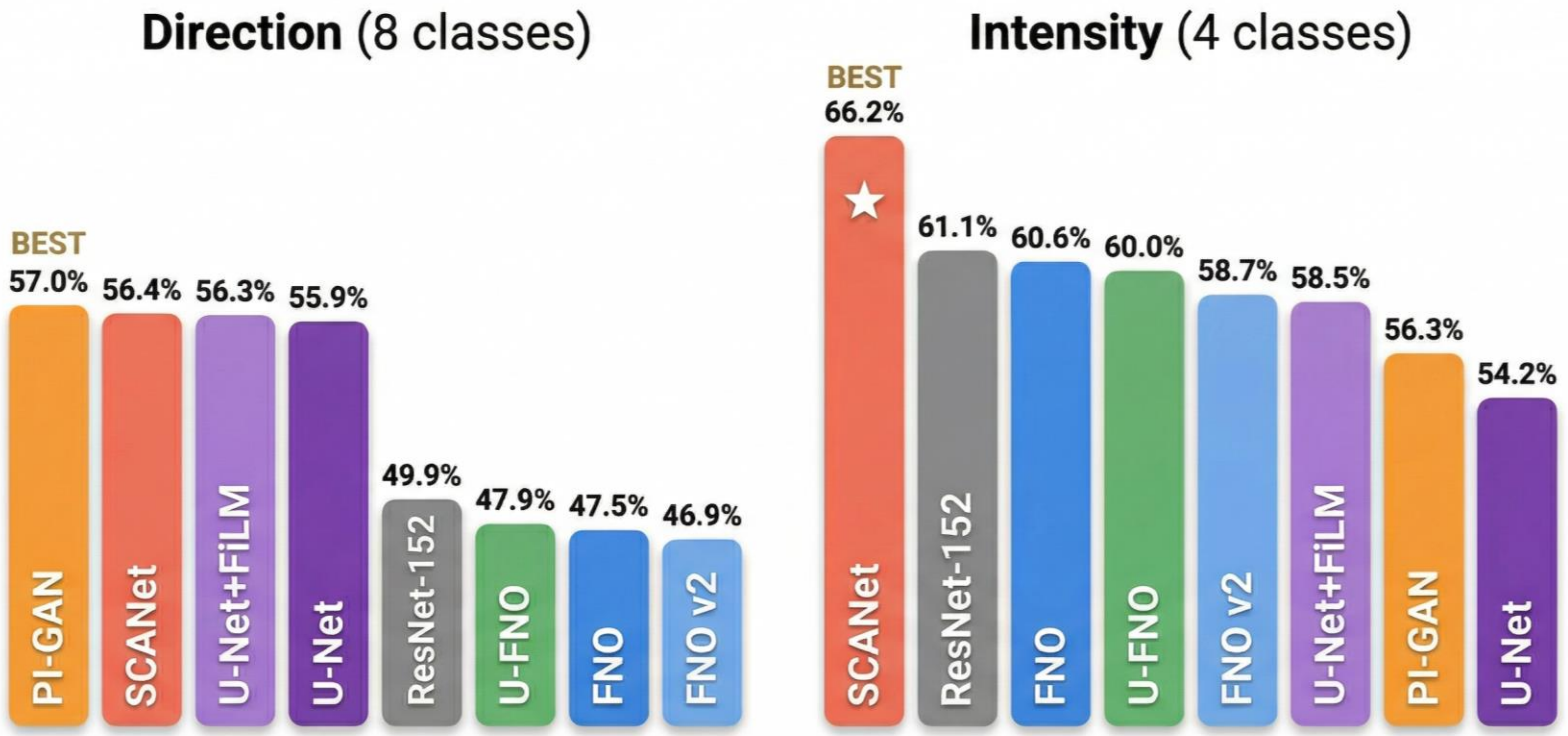
Learns the optimal spectral/spatial mix

# U-FNO: Learned Gating



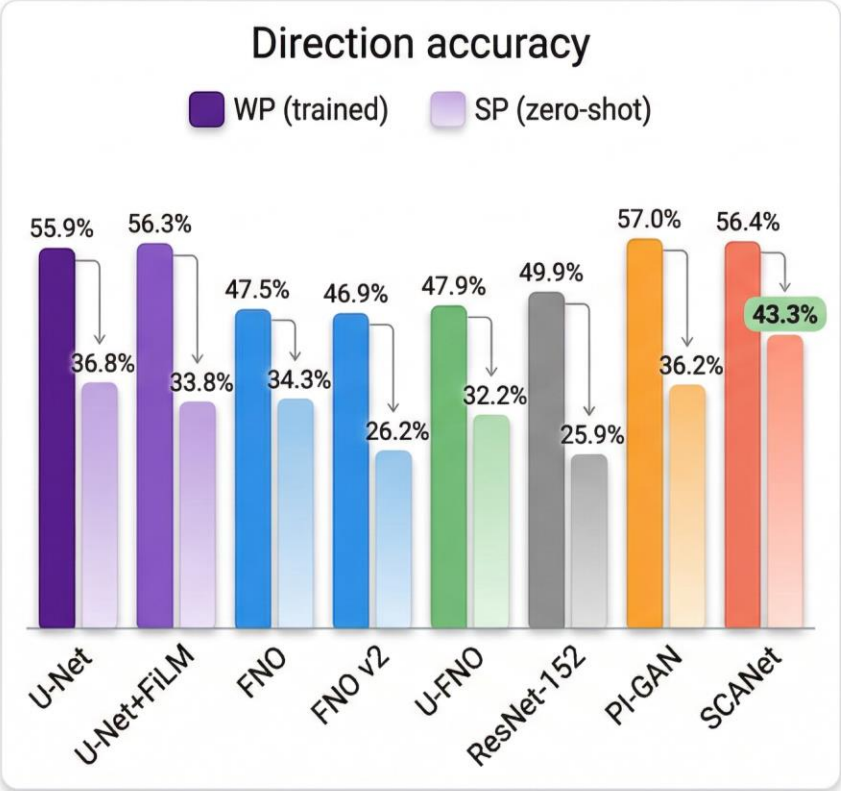
# Results and Analysis

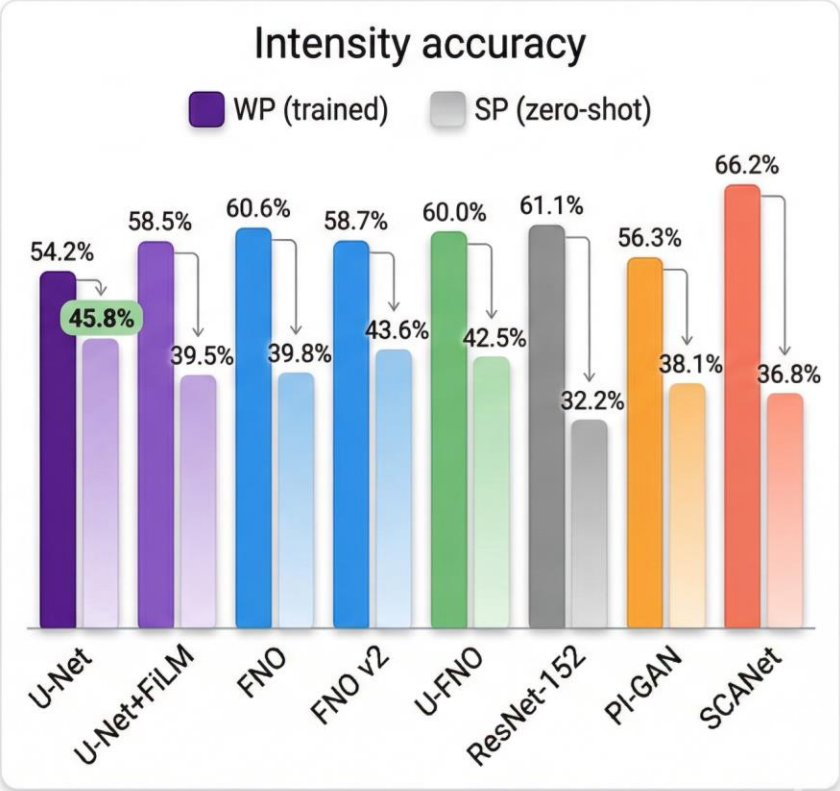
# WP In-Basin Performance (% Accuracy)




SCANet dominates intensity (+5pp over #2) while PI-GAN leads direction

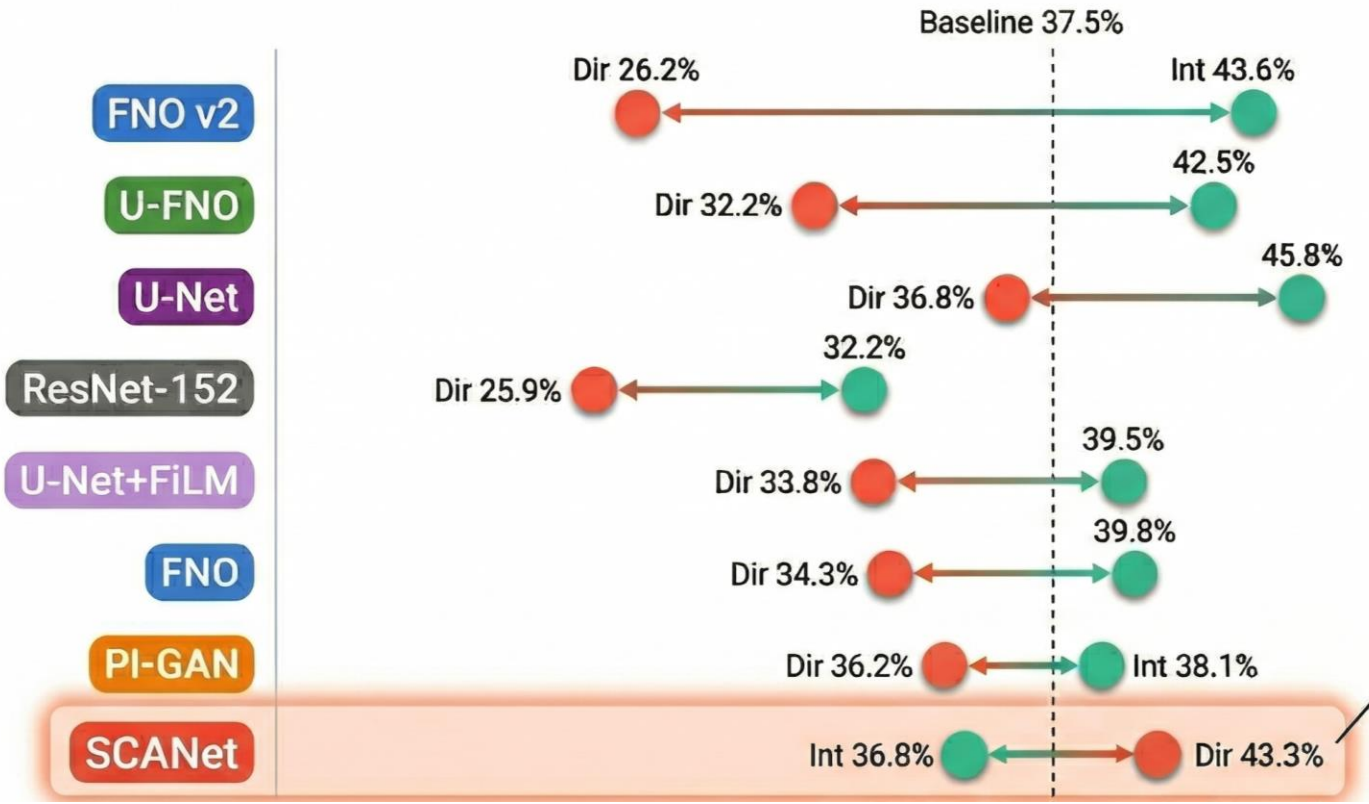
# Zero-Shot Transfer (% Accuracy)



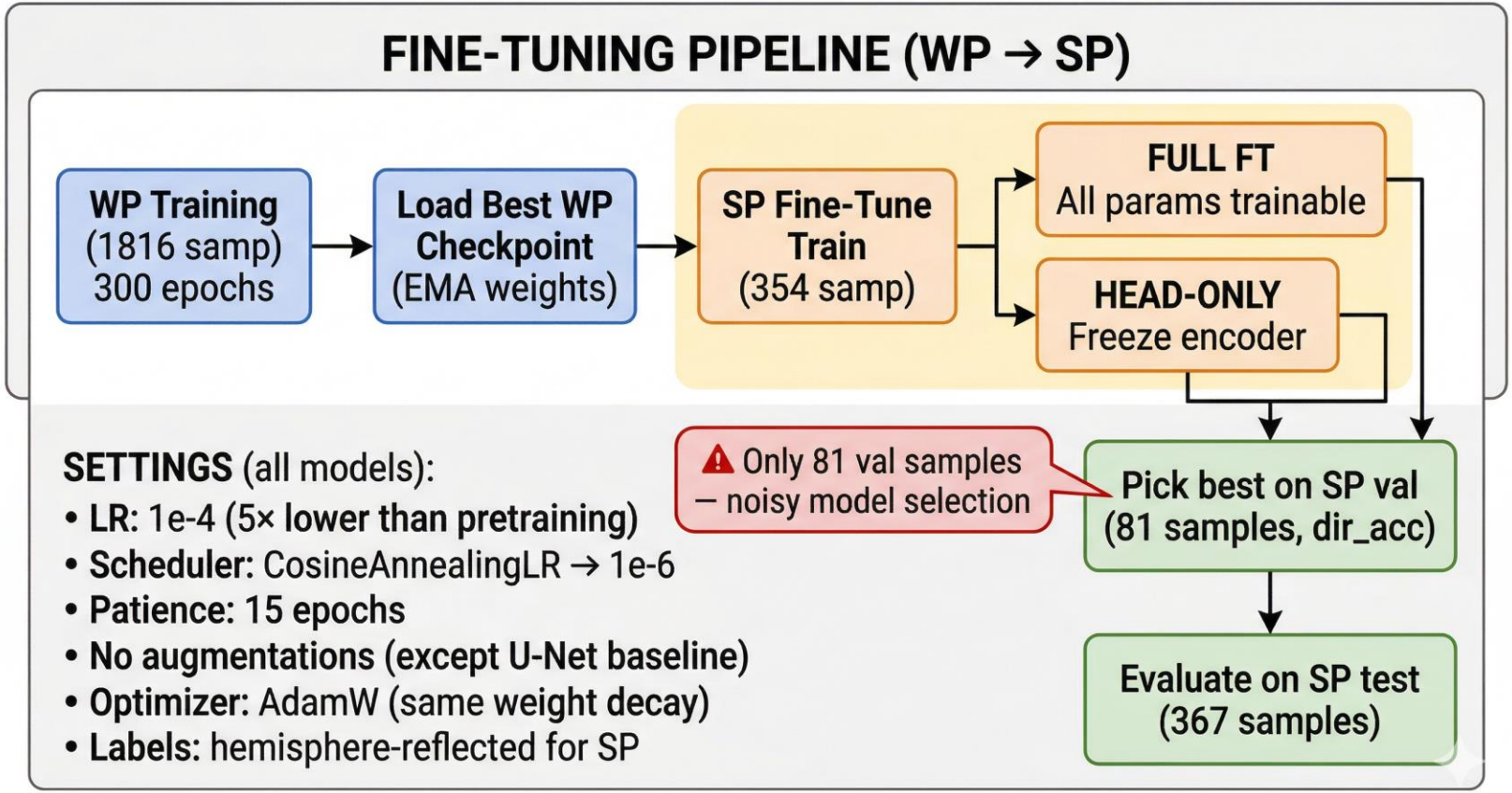

SCANet retains the most direction knowledge; simple U-Net is surprisingly robust for intensity

# Direction vs Intensity Transfer Gap



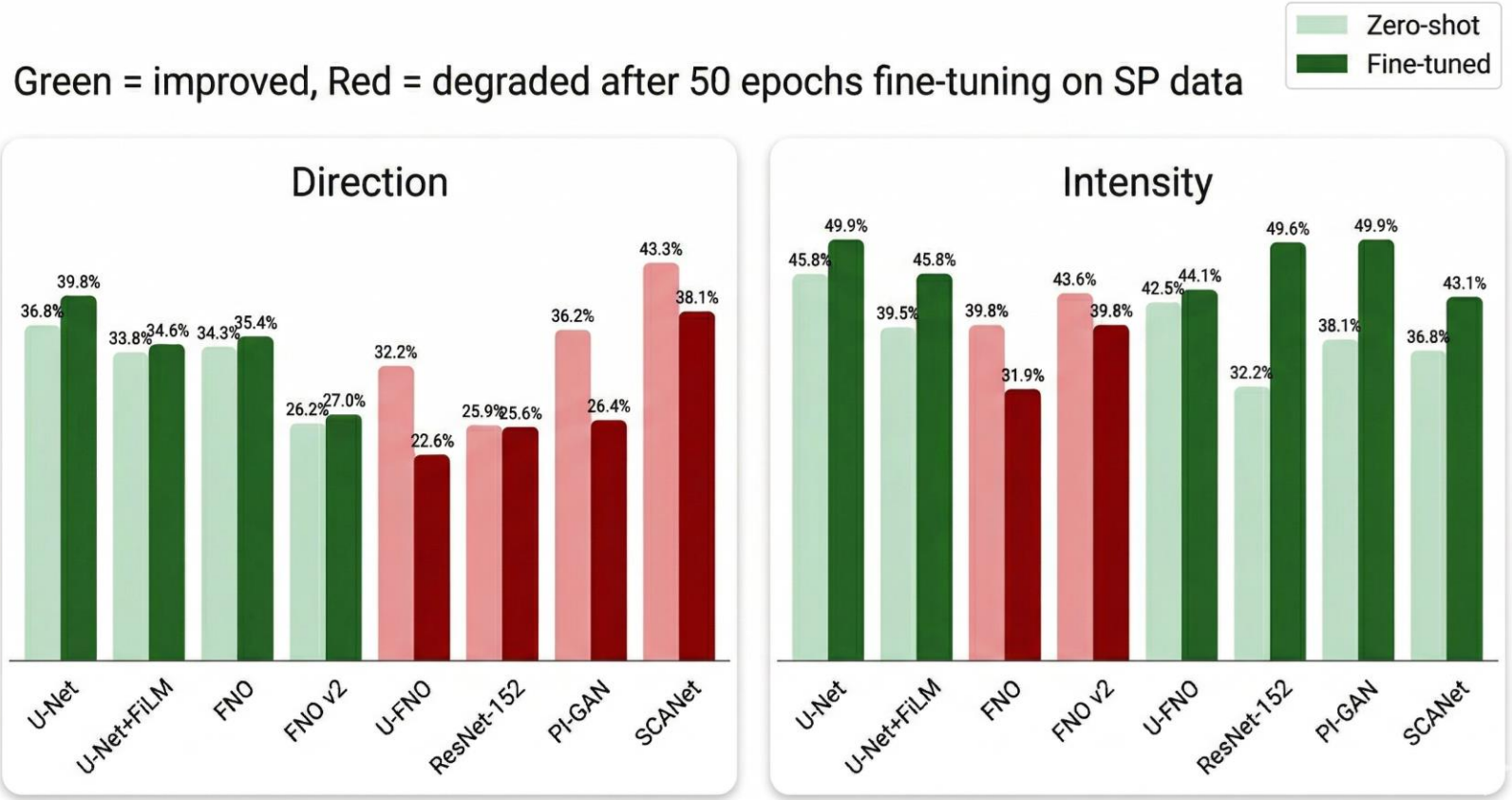

SCANet is the **ONLY** model where direction > **intensity in zero-shot** — its cross-attention captures transferable directional patterns

# Fine-Tuning: Double-Edged Sword I

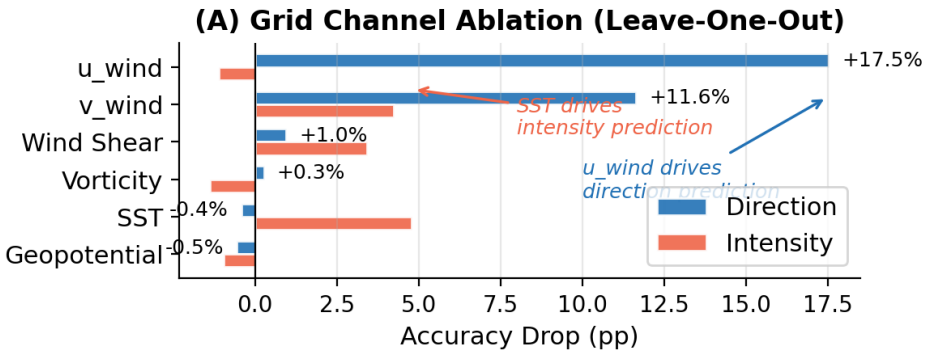


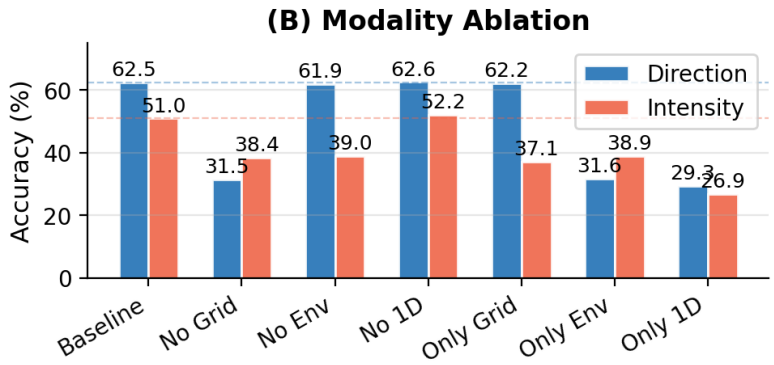
# Fine-Tuning: Double-Edged Sword II

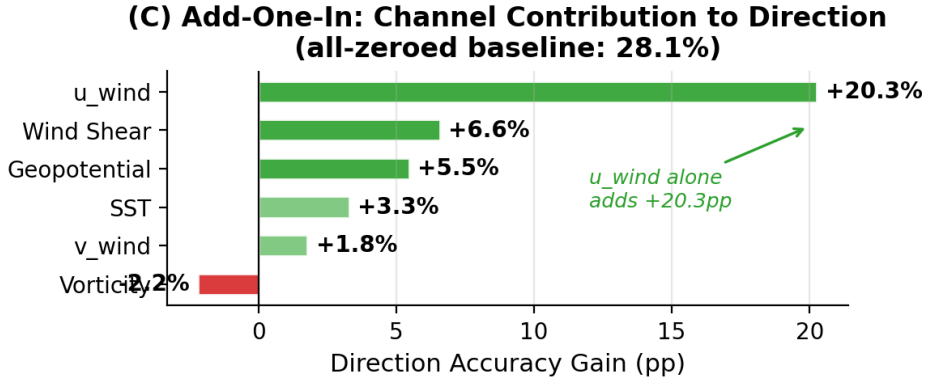
Green = improved, Red = degraded after 50 epochs fine-tuning on SP data

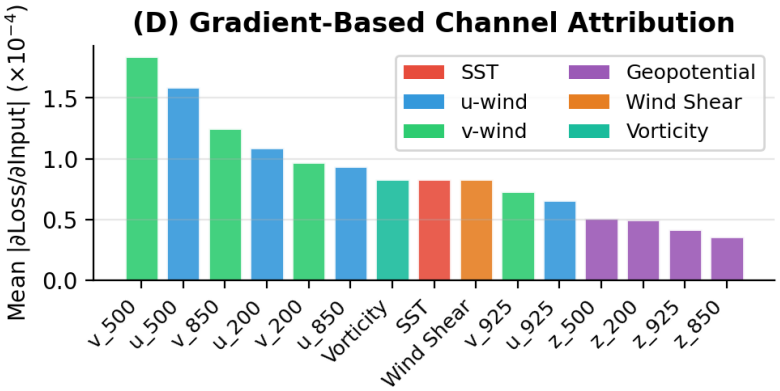


# Ablation Analysis: U-Net Baseline WP Val

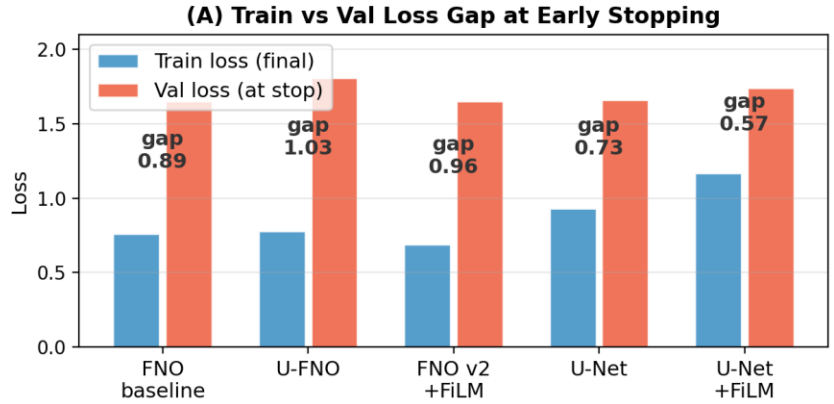


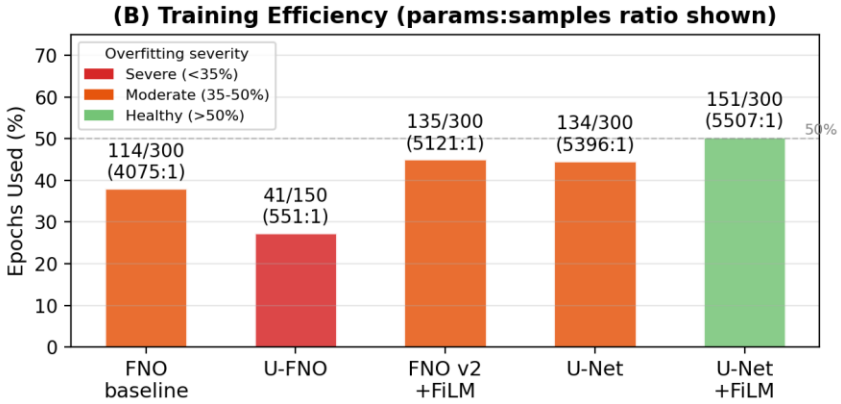



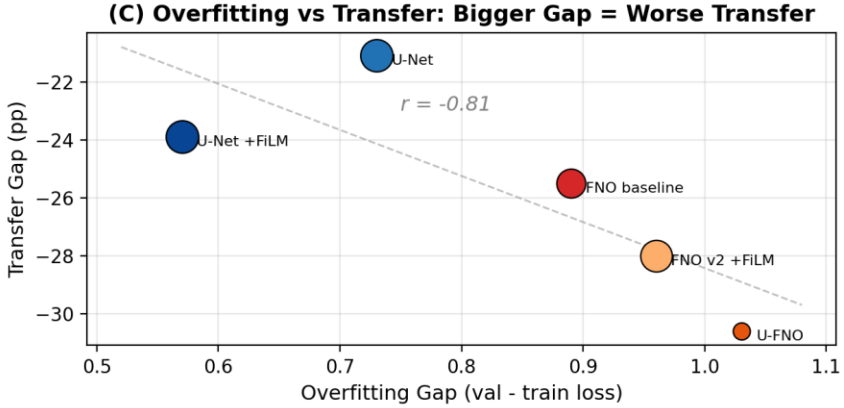



# Overfitting Analysis: Data-Limited Regime




**(D) Summary: Overfitting & Generalisation**

| Model | Params | Ratio | Epochs | Overfit Gap | WP Dir | SP Dir |
|---|---|---|---|---|---|---|
| FNO | 7.4M | 4,074:1 | 114/300 | 0.89 | 50.3% | 24.8% |
| U-FNO | 1.0M | 551:1 | 41/150 | 1.03 | 50.8% | 20.2% |
| FNO v2 | 9.3M | 5,121:1 | 135/300 | 0.96 | 49.3% | 21.3% |
| U-Net | 9.8M | 5,396:1 | 134/300 | 0.73 | 62.5% | 41.4% |
| U-Net+FiLM | 10.0M | 5,507:1 | 151/300 | 0.57 | 62.9% | 39.0% |

*Green = low gap, Red = high. All models trained on 1816 WP samples.*

# Key Takeaways & Future Work

✓ **Hypothesis supported**

✓ **No single architecture** excels at both direction & intensity classification

✓ Cross-basin transfer is viable for **intensity** with minimal fine-tuning.
✓ **Direction** requires more target data or explicit domain adaptation.

## Future work

1. Test on other basin pairs (e.g., **North & South Indian**)
2. Two additional architectures to be further explored **PI-GAN** & **SCANet**

# Thank You!

Questions?

# References


[1] Imperial College London, Grantham Institute. "Modelling Tropical Cyclones." Available at: https://www.imperial.ac.uk/grantham/research/climate-science/modelling-tropical-cyclones/

[2] Huang, C. et al. (2025) "Benchmark dataset and deep learning method for global tropical cyclone forecasting." Nature Communications, 16, 5923.

[3] DeMaria, M. & Kaplan, J. (1994) "A Statistical Hurricane Intensity Prediction Scheme (SHIPS)." Weather and Forecasting, 9(2), 209–220.

[4] Li, Z. et al. (2021) "Fourier Neural Operator for Parametric Partial Differential Equations." ICLR 2021.

[5] Perez, E. et al. (2018) "FiLM: Visual Reasoning with a General Conditioning Layer." AAAI 2018, pp. 3942-3951.

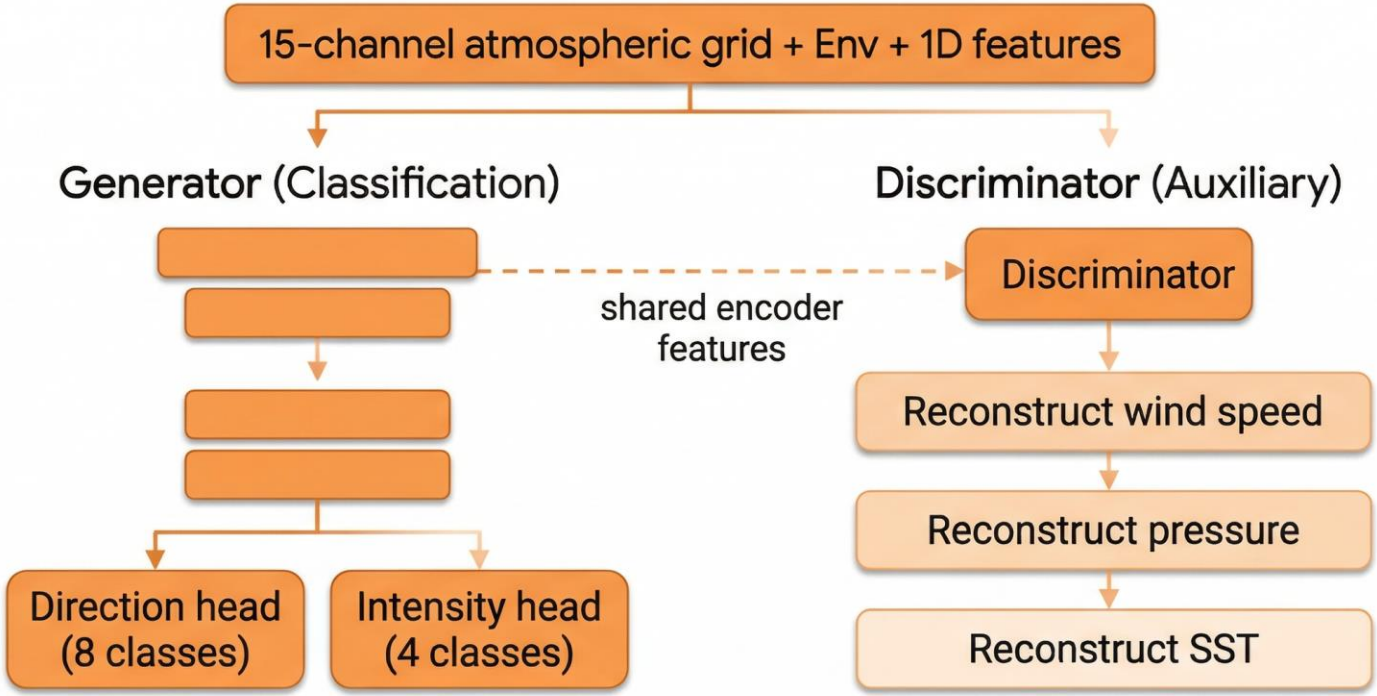
[6] Wen, G. et al. (2022) "U-FNO — An enhanced Fourier neural operator-based deep-learning model for multiphase flow." Advances in Water Resources, 163, 104180.

[7] Raissi, M. et al. (2019) "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear PDEs." Journal of Computational Physics, 378, 686-707.

[8] Vaswani, A. et al. (2017) "Attention Is All You Need." NeurIPS 2017, pp. 5998-6008.

# Appendix

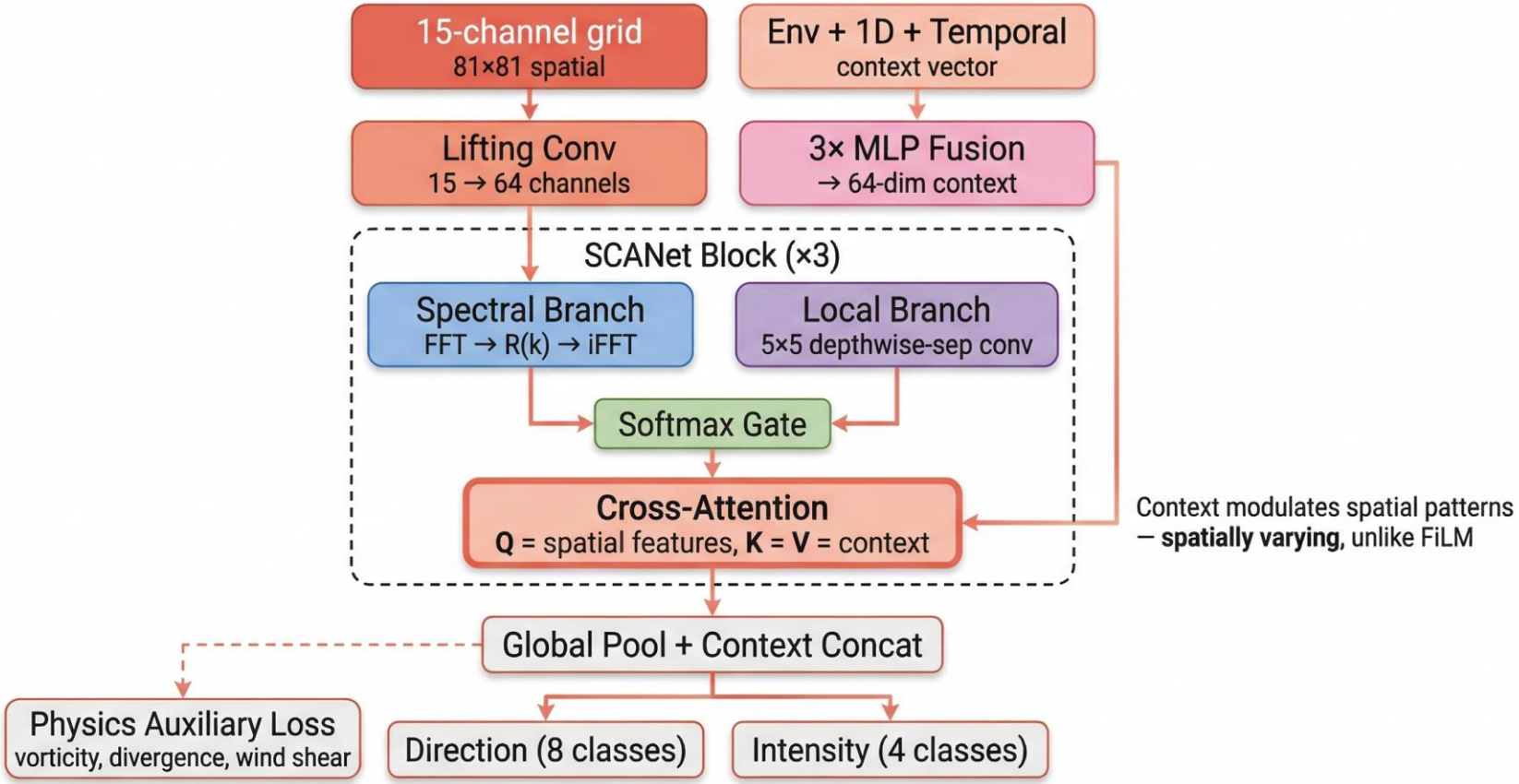
# PI-GAN: Physics-Informed GAN




Total Loss = Classification Loss + $\lambda$ · Physics Reconstruction Loss

Auxiliary physics losses act as regularisation — the model must understand physical variables to classify well

# **SCANet**: Spectral Cross-Attention Network




Cross-attention lets environmental context guide spatial feature extraction — captures transferable directional patterns